MILITARY SPECIFICATION

WIRING HARNESS, RELAY ASSEMBLY, M75

1. SCOPE

1.1 This specification covers requirements, quality assurance provisions, and the preparation for delivery criteria for one type of wiring harness, entitled Wiring Harness, Relay Assembly, M75.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-S-4456	- Shocks, Variable Duration, Method and Apparatus For.

STANDARDS

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-810	- Environmental Test Methods.
MIL-STD-1167	- Ammunition Data Card.
MIL-STD-1168	- Lot Numbering of Ammunition.
MIL-STD-1169	- Packaging, Packing and Marking for Shipment of Inert Ammunition Components.
MIL-STD-1460	- Soldering of Electrical Connections and Printed Wiring Assemblies, Procedure For.

DRAWINGS

ORDNANCE CORPS

8825023	- Wiring Harness, Relay Assembly, M-75.
EL-8884687	- Equipment List.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material complies with the specifications listed on the applicable drawing.

3.2 Assembly. The harness shall comply with all requirements specified on the drawing (dwg) 8825023, with all requirements specified herein, and with the requirements of all applicable specifications to the extent specified on the drawing and herein.

3.3 First article testing. Requirements for the submission of first article samples by the contractor shall be as specified in the contract (see 4.2).

3.4 Serialization. A six letter serial number shall be assigned to the item. Serial numbers required on these items shall be obtained by means of a letter addressed to: Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTENTION: SARPA-QA-T-M. The letter of request shall include:

a. Exact nomenclature of the item as specified on the applicable drawing.
b. Name of the contractor.
c. Contract number or procurement request order number.
d. Number of serial numbers required.

3.5 Circuit and contact resistance. The resistance of each circuit involving relay contacts together with the connectors associated with these circuits and the resistance of all other connectors including their circuits shall not exceed the values specified in Table V when tested as specified in 4.4.1.

3.6 Insulation resistance. The insulation resistance between electrically isolated circuits and between ungrounded circuits shall be not less than 100 megohms for ambient conditions but shall not be less than 20 megohms during the altitude test when subjected to a direct current (dc) potential of 420 plus or minus (±) 20 volts and capacitors C101 and C201 shall show no evidence of failure when tested as specified in Table VII in 4.4.2.

3.7 Environmental testing. The harness shall comply with MIL-STD-810, visual inspection and failure criteria, section 3. The harness shall also comply with 4.3.3.1 before and after each environmental test. Testing shall be as specified in 4.4.3. Tests shall be conducted in the order specified.

3.7.1 Shock (spares only). The wiring harness, relay assembly shall withstand impact shocks of 50 ± 0.5 gravity units (g's) along the principal axes without evidence of damage and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.1.

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3.7.2 Vibration (spares only). The wiring harness, relay assembly shall withstand vibration at the frequencies, amplitude inches or acceleration in g's for the time specified in Table I in each of its three mutually perpendicular axes, without evidence of damage such as loosened screws, loose internal components or contact chatter in excess of 10 microseconds and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.2.

Table I

Frequency (Hz)	Time of Cycle (Min)	Acceleration (g's)	Amplitude Inches	Condition	No. of Cycles in Each Axis
5 - 55	2.5		0.02	All relays degenerized.	1
55 - 500	5.0	10 ± 0.2		K102 and K202 energized all others de-energized.	1
500 - 55	5.0	10 ± 0.2		All relays de-energized.	1
55 - 5	2.5		0.02	K102 and K202 energized all others de-energized.	1

3.7.3 Altitude. The wiring harness, relay assembly shall withstand simultaneously, exposure to a pressure of 5.54 ± 0.5 inches of mercury (corresponding to an altitude of 40,000 feet) and a temperature of +165 ± 2 degrees Fahrenheit (°F.). The wiring harness relay assembly shall also withstand simultaneously, exposure to a pressure 5.54 ± 0.5 inches of mercury and a temperature of -65 ± 2°F. The assembly shall exhibit no evidence of damage such as deterioration of the pigtail insulation sleeving or deterioration of the internal wiring insulation and shall comply with the requirements of 3.5 and 3.6 after being tested as specified in 4.4.3.3.

3.8 Soldering. All soldering shall comply with the requirements specified in MIL-STD-1460.

3.9 Workmanship.

3.9.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.9.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.9.3 Burrs. Parts shall be free from burrs.

3.9.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

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3.9.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agent.

3.9.6 Electrical. Cable assemblies and wires shall be dressed in such a manner as to prevent pinching or chafing of the wire insulation. Ends of wires shall be free from burrs, sharp edges, or points. All parts shall be free from flux, chips, dirt, grease, rust or foreign material.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection.

4.2.1 Quantity to be submitted. When the contract requires first article inspection, the contractor shall submit 3 complete assemblies (see dwg 8825023) for first article testing to the testing facility designated in the invitation for bids or request for proposal.

4.2.2 Inspection and testing to be performed. The first article inspection sample shall be subjected to the inspections and tests specified in Table II. Tests shall be conducted in order specified in Table II.

Table II

First article inspection and tests

Inspection or Test	Dest	Non-Dest
1. Classification of characteristics (see 4.3.2.1)		X
2. Circuit and contact resistance (see 3.5)		X
3. Insulation resistance (see 3.6)		X
4. Shock (see 3.7.1)	X	
5. Vibration (see 3.7.2)	X	
6. Altitude (see 3.7.3)		X

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4.2.3 Inspection equipment. The inspection and testing specified in 4.2.2 shall be accomplished by using the gages and test equipment specified in 4.3.4.1.

4.2.4 Rejection. If any item fails to comply with any of the applicable requirements, the first article sample shall be rejected.

4.3 Quality conformance inspection.

4.3.1 Lot formation.

4.3.1.1 Production lot. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168.

4.3.1.2 Inspection lot. A lot or batch and its formation, size, and presentation is described in MIL-STD-105, Section 5. Accordingly, a lot shall mean an inspection lot and a batch shall mean an inspection batch for the purpose of this specification. The manner in which each inspection lot or batch is to be presented and identified by the contractor, shall be designated or approved by the Government representative.

4.3.2 Product inspection examination. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table (see Table III and Table IV), and inspected for all characteristics in the applicable classification. Any characteristics in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Table III
Major Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-72	all	140-141	124	256-259	177	541-549	230
73	72	142	125	260-262	178	550-559	231
74	73	143-144	126	263-265	179	560-569	232
75	74	145	127	266-268	180	570-579	233
76	75	146-147	128	269-271	181	580-590	234
77	76	148-149	129	272-275	182	591-601	235
78	77	150	130	276-278	183	602-613	236
79	78	151-152	131	279-282	184	614-625	237
80	79	153-154	132	283-285	185	626-637	238
81-82	80	155-156	133	286-289	186	638-649	239
83	81	157	134	290-293	187	650-662	240
84	82	158-159	135	294-296	188	663-676	241
85	83	160-161	136	297-300	189	677-690	242
86	84	162-163	137	301-304	190	691-704	243
87	85	164-165	138	305-308	191	705-719	244
88-89	86	166-167	139	309-312	192	720-735	245
90	87	168	140	313-316	193	736-751	246
91	88	169-170	141	317-320	194	752-768	247
92	89	171-172	142	321-324	195	769-785	248
93	90	173-174	143	325-329	196	786-804	249
94-95	91	175-176	144	330-333	197	805-823	250
96	92	177-178	145	334-338	198	824-842	251
97	93	179-180	146	339-342	199	843-863	252
98	94	181-182	147	343-347	200	864-884	253
99-100	95	183-184	148	348-352	201	885-907	254
101	96	185-187	149	353-356	202	908-930	255
102	97	188-189	150	357-361	203	931-955	256
103	98	190-191	151	362-366	204	956-981	257
104-105	99	192-193	152	367-372	205	982-1008	258
106	100	194-195	153	373-377	206	1009-1036	259
107	101	196-197	154	378-382	207	1037-1066	260
108-109	102	198-200	155	383-388	208	1067-1098	261
110	103	201-202	156	389-393	209	1099-1131	262
111	104	203-204	157	394-399	210	1132-1166	263
112-113	105	205-207	158	400-405	211	1167-1204	264
114	106	208-209	159	406-411	212	1205-1243	265
115	107	210-212	160	412-417	213	1244-1285	266
116-117	108	213-214	161	418-424	214	1286-1330	267
118	109	215-217	162	425-430	215	1331-1378	268
119	110	218-219	163	431-437	216	1379-1429	269
120-121	111	220-222	164	438-443	217	1430-1483	270
122	112	223-224	165	444-450	218	1484-1542	271
123-124	113	225-227	166	451-457	219	1543-1605	272
125	114	228-230	167	458-465	220	1606-1673	273
126-127	115	231-232	168	466-472	221	1674-1747	274
128	116	233-235	169	473-480	222	1748-1827	275
129-130	117	236-238	170	481-488	223	1828-1914	276
131	118	239-241	171	489-496	224	1915-2010	277
132-133	119	242-243	172	497-504	225	2011-2115	278
134	120	244-246	173	505-513	226	2116-2232	279
135-136	121	247-249	174	514-521	227	2233-2361	280
137	122	250-252	175	522-530	228	over 2361	298
138-139	123	253-255	176	531-540	229		

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Table IV
Minor Characteristics

Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size	Lot Size	Sample Size
1-16	all	28-29	23	52-56	31	132-156	39
17	16	30-31	24	57-62	32	157-191	40
18	17	32-34	25	63-69	33	192-244	41
19-20	18	35-36	26	70-76	34	245-334	42
21	19	37-40	27	77-86	35	335-519	43
22-23	20	41-43	28	87-97	36	520-1120	44
24-25	21	44-47	29	98-112	37	over 1120	45
26-27	22	48-51	30	113-131	38		

4.3.2.1 Classification of characteristics (see 3.2). Critical, major and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Wiring harness, relay assembly (see dwg 8825023).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	
Major:		
101.	Foreign matter on inserts unless removable	Visual
102.	Any pin bent	Visual
103.	Foreign matter on pin	Visual
104.	Any pin missing (as evidenced by empty hole in insert)	Visual
105.	Improperly soldered joint (see 3.8)	Visual
106.	Solder loose on terminal (see 3.8)	Visual
107.	Deformation of connector, bracket, or adapter	Visual
108.	Crack in any component	Visual
109.	Terminal board loose or missing	Manual/Visual
110.	Any component missing or insecurely assembled	Manual/Visual
111.	Any required stamping or marking mislocated, misleading or unidentifiable (incorrect, incomplete, superimposed or smeared)	Visual
112.	Keyway missing or mislocated	Visual
113.	Chip or gouge extending into pin hole	Visual
114.	Portion of connector, adapter or bracket broken away (minor chips, scratches, burrs, or gouges which do not affect mating are not objectional)	Visual

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115.	Serialization number missing or improperly assigned (see 3.4)	Visual
116.	Foreign matter	Visual
*117.	Height of pin(s) (pins of the same diameter must all appear to be within 1/16 inch in height relative to the adjacent pins)	Visual/Gage
**118.	Depth of socket(s) (female sockets of the same diameter must all appear to be within 1/16 inch in depth relative to the adjacent socket)	Visual/Gage
119.	Objective evidence relative to material missing or improper (see 3.1)	Visual
Minor:		
201.	Evidence of poor workmanship, except as otherwise specified herein (see 3.9)	Visual

*In the event pin(s) do not appear to be within 1/16 inch in height relative to the adjacent pins, a gage (see 4.3.4.1) shall be used to determine the variation from the normal height. Reject any pin that varies by more than 0.063 inch maximum from the normal pin height of the individual connector. The normal height of the individual connectors pins shall be determined by calculating the average height from measurements taken of pins adjacent to the pin that appears to be defective. The forward metal edge of the connector shall be utilized as the reference surface for all measurements of pin heights.

**In the event socket inserts do not appear to be within 1/16 inch in depth relative to the adjacent sockets, a gage (see 4.3.4.1) shall be used to determine the variation from the normal depth. Reject any insert that varies by more than 0.063 inch maximum from the normal socket insert depth of the individual connector. The normal depth of the individual connector sockets shall be determined by calculating the average depth from the measurements taken of the sockets adjacent to the socket that appears to be defective. The forward edge of the connector insert shall be utilized as the reference surface for all measurements of socket depths.

4.3.2.1.2 Preparation for delivery (see 5).

Categories	Characteristics	Method of Inspection
Critical:	None defined.	

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Major:

101. Preservation and packaging not in accordance with MIL-STD-1169.....Visual
102. Packing not in accordance with MIL-STD-1169.....Visual
103. Marking not in accordance with MIL-STD-1169.....Visual

Minor:

201. Evidence of poor workmanship (see 3.9).....Visual

4.3.3 Testing. All tests shall be performed. Tests shall be performed in the order specified below:

Test	Test Classification
a. Circuit and contact resistance	Major
b. Insulation resistance	Major
c. Shock (spares only)	Major
d. Vibration (spares only)	Major
e. Altitude	Major

Of these tests, c and d are considered destructive and samples so tested shall not be returned to the lot. Destructive samples are over and above the deliverable quantity of items required by the contract.

4.3.3.1 Circuit and contact resistance, and insulation resistance. Each harness shall be tested in accordance with 4.4.1 and 4.4.2. If any item fails to comply with 3.5 and 3.6, the item shall be rejected and removed from the lot.

4.3.3.2 Shock (spares only), vibration (spares only), and altitude. One sample harness shall be selected from each months production and subjected to these tests in the order specified (see 4.4.3.1 through 4.4.3.3). If any harness fails to comply with the applicable requirements (see 3.7.1 through 3.7.3), the lot shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL) (see EL-8884687). When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment and obtain approval prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired (see 6.3).

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b. The code "CD" is listed in the "Part or Identifying Number" column on the EL (see 6.1.2).

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on ELs. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Circuit and contact resistance. The resistance of each circuit involving relay contacts together with the connectors associated with these circuits and the resistance of all other circuits including their connectors shall be tested at 100 ± 2 milliamperes. The applied voltage shall not exceed 28 volts dc. The values of resistance for the various circuits shall be as specified in Table V. Relay coil resistance measurements should be made prior to resistance testing relay contacts. Relay contacts shall be tested on initial closure.

Table V

To Energize	Apply Rated Current At	From	To
K101	28 ± 2 Volts dc	J8-N	Ground
K102	15 ± 2 Volts dc	J8-M	Ground
K103	28 ± 2 Volts dc	J8-X	J8-B
K104	28 ± 2 Volts dc	J6-f	J2-A
K105	28 ± 2 Volts dc	J9-h	Ground
K201	28 ± 2 Volts dc	J8-W	Ground
K202	15 ± 2 Volts dc	J2-f	J8-M
K203	28 ± 2 Volts dc	J8-G	J8-E
K204	28 ± 2 Volts dc	J7-f	J2-c
K205	28 ± 2 Volts dc	J9-Y	Ground

From	To	Ohms **	From	To	Ohms **
*J2-K	J9-B	12 ± .6	*P2-F	P2-U	Open
*J2-Y	J9-K	12 ± .6	*P1-F	P1-U	Open
*J8-N	J3-B	50 ± 2	J7-M	P2-A	0.1
*J8-W	J3-D	50.6 ± 1.4	J7-L	P2-H	0.1
*J2-f	J8-M	66.2 ± 6.6	P1-S	J6-A	0.1
*J3-c	J3-N	66.2 ± 6.6	J7-C	P1-T	0.1

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From	To	Ohms **
*J6-f	J2-A	220 ± 17
*J7-f	J2-c	220 ± 17
*J2-g	J8-B	220 ± 17
*J3-H	J8-E	220 ± 17
*J9-h	J9-0	220 ± 17
*J9-Y	J9-0	220 ± 17
*J8-N	J8-C	220 ± 17
*J8 W	J8-C	220 ± 17
J6-B	P1-E	0.1
P1-J	J6-C	0.1
P1-A	J7-g	0.1
J8-W	P4-D	0.1
J8-N	P3-H	0.1
J5-M	P2-P	0.1
J5-C	P1-V	0.1
J9-h	P3-C	0.1
J9-Y	P4-C	0.1
P3-K	P4-K	0.1
J7-U	P4-E	0.1
J1-T	P2-V	0.1
J6-f	P2-C	0.1
J7-e	P4-L	0.1
J9-H	P4-J	0.1
J5-M	P2-L	0.1
J5-C	P1-W	0.1
J9-h	P3-F	0.1
J9-Y	P4-F	0.1
J9-J	P1-L	0.1
P2-B	P3-J	0.1
J1-T	P2-M	0.1
J2-a	P1-P	0.1
J6-e	P3-E	0.1
J6-U	P3-L	0.1
J5-M	P2-D	0.1
J5-C	P1-C	0.1
P1-X	P2-X	0.1
J1-G	P4-H	0.1
J9-J	P1-N	0.1
J1-W	P3-D	0.1
J7-f	P2-W	0.1
J2-a	P1-D	0.1
J7-Y	P1-B	0.1
J7-J	P2-E	0.1
J5-M	P2-N	0.1
J5-C	P1-M	0.1
J8-N	P3-D	0.06 Eng. K101
J8-W	P4-H	0.06 Eng. K101
*J1-J	J5-P	0.08 Eng. K101
*J1-K	J5-T	0.08 Eng. K101
J2-d	J5-H	0.1

From	To	Ohms **
J6-g	P1-H	0.1
J6-K	P2-K	0.1
J7-K	P2-J	0.1
P1-K	J7-B	0.1
J6-L	P2-T	0.1
J6-J	P2-R	0.1
P1-R	J7-A	0.1
J6-M	P2-S	0.1
J1-G	J1-H	0.1
J2-D	J8-E	0.1
J9-J	J9-0	0.1
J6-M	J8-L	0.1
J7-K	J8-a	0.1
*J8-N	P3-D	0.06 Eng. K201
*J8-W	P4-H	0.06 Eng. K201
*J1-J	J5-P	0.08 Eng. K201
*J1-K	J5-T	0.08 Eng. K201
J2-d	J7-E	0.1
J1-G	J3-E	0.1
J2-D	J7-T	0.1
J3-C	J8-M	0.1
J6-K	J8-J	0.1
J6-L	J8-K	0.1
J9-h	P3-D	0.06 Eng. K105
*J2-S	J8-X	0.06 Eng. K105
*J2-D	J2-T	0.06 Eng. K105
J1-M	J6-R	0.1
J2-g	J6-T	0.1
J1-W	J1-X	0.1
J3-C	J7-b	0.1
J6-J	J8-H	0.1
J7-h	J8-Y	0.1
J9-Y	P4-H	0.08 Eng. K205
*J2-S	J8-X	0.06 Eng. K205
*J2-D	J2-T	0.06 Eng. K205
J3-H	J8-G	0.1
J2-g	J8-X	0.1
J1-W	J3-F	0.1
*J3-C	J7-X	0.08
J7-J	J8-Z	0.1
*J7-h	J7-R	0.12
J7-D	J2-C	0.1
J3-L	J8-B	0.1
J2-M	J6-c	0.1
J2-N	J6-b	0.1
*J9-h	J9-M	0.09 Eng. K102
*J2-g	P3-H	0.08 Eng. K102
*J8-W	J8-E	0.08 Eng. K102
*J2-a	J7-X	0.14
J7-M	J8-T	0.1

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J7-Y	J8-R	0.1		
J2-R	J4-D	0.1		
J2-J	J5-F	0.1		
J2-e	J4-B	0.1		
*J9-Y	J9-e	0.09	Eng.	K202
*J2-k	P3-H	0.08	Eng.	K202
*J8-W	J2-j	0.06	Eng.	K202
J7-L	J8-S	0.1		
J2-R	J5-D	0.1		
J2-J	J6-G	0.1		
J2-e	J5-K	0.1		
*J2-J	J6-c	0.06	Eng.	K103
*J2-N	J7-E	0.08	Eng.	K103
*J3-L	J3-N	0.08	Eng.	K103
J7-V	J3-D	0.1		
J2-R	J6-d	0.1		
J6-V	J3-B	0.1		
J2-e	J7-d	0.1		
*J2-J	J6-c	0.08	Eng.	K203
*J2-N	J7-E	0.06	Eng.	K203
*J3-H	J3-N	0.08	Eng.	K203
*J2-e	J2-U	0.06	Eng.	K104
*J1-M	J2-C	0.12	Eng.	K104
*J2-R	J2-U	0.06	Eng.	K104
*J2-e	J2-U	0.06	Eng.	K204
*J1-M	J2-C	0.08	Eng.	K204
*J2-R	J2-U	0.06	Eng.	K204
J1-T	J1-U	0.1		
J1-T	J2-U	0.1		
J1-T	J3-N	0.1		
J1-T	J4-A	0.1		
J1-T	J4-C	0.1		
J1-T	J5-E	0.1		
J1-T	J5-G	0.1		
J1-T	J5-J	0.1		
J1-T	J5-S	0.1		
J1-T	J6-E	0.1		
J1-T	J7-R	0.1		
J1-T	J8-C	0.1		
J1-T	J8-V	0.1		
J1-T	J9-0	0.1		
J1-T	J9-P	0.1		
*J2-F	J9-L	0.11		
*J2-F	J6-m	0.3		
J2-W	J7-1	0.1		
*J2-W	J9-G	0.11		
*J2-W	J7-m	0.3		
J2-P	J9-U	0.1		
J2-h	J9-V	0.1		
J2-O	J6-X	0.1		
J2-V	J6-h	0.1		

J1-T	J9-R	0.1
J1-T	J9-T	0.1
J1-T	J9-g	0.1
*J1-T	Ground	0.1
J1-L	J2-G	0.1
J1-L	J3-J	0.1
J1-N	J1-P	0.1
J1-N	J3-N	0.1
J1-a	J2-b	0.1
J1-a	J7-a	0.1
J1-D	J2-I	0.1
J1-D	J6-a	0.1
J1-A	J1-R	0.1
J1-A	J6-F	0.1
J1-C	J3-I	0.1
J1-C	J7-S	0.1
J1-E	J1-V	0.1
J1-E	J3-O	0.1
J1-B	J3-K	0.1
J1-B	J6-S	0.1
J1-d	J5-L	0.1
J1-b	J5-N	0.1
J1-Z	J2-H	0.1
J1-Z	J2-B	0.15
J1-F	J6-Y	0.1
J1-F	J9-C	0.1
J1-S	J3-P	0.1
J1-S	J7-G	0.1
J1-Y	J8-P	0.1
J2-L	J6-P	0.1
*J2-L	J9-c	0.06
*J2-L	J6-N	0.3
J2-Z	J7-P	0.1
*J2-Z	J9-N	0.06
*J2-Z	J7-N	0.3
J2-E	J6-j	0.1
*J2-E	J9-D	0.13
*J2-E	J6-k	0.3
J2-X	J7-j	0.1
*J2-X	J9-I	0.3
*J2-X	J7-k	0.13
J2-F	J6-1	0.1
J6-W	J9-f	0.1
J6-H	J9-d	0.1
J6-Z	J8-U	0.1
J6-Z	J9-b	0.1
J7-H	J9-A	0.1
J7-F	J9-F	0.1
J7-Z	J8-D	0.1
J7-Z	J9-k	0.1
J7-c	J9-E	0.1

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J3-G	J8-A	0.1	J7-W	J8-b	0.1
J3-A	J8-F	0.1	J7-W	J9-j	0.1
J5-R	J6-D	0.1			

*Readings for these circuits only shall be forwarded to Picatinny Arsenal (see 5.4).

**If a tolerance is not specified the value given is the maximum value allowed.

4.4.2 Insulation resistance. Each connector contract, except the circuits involving capacitors, shall be subjected to the required dc potential (see 3.6) until a steady reading is obtained. The proper contacts and values shall be as specified in Table VI. The circuits involving capacitors shall be tested as specified in Table VII.

Table VI

Pin Energized	All Other Pins to Ground Except	Megohms (min.)
J1-L	J2-G, and J3-J	100
J1-W	J1-X, J3-F and P3-D	100
J1-N	J1-P and J3-M	100
J1-a	J2-b and J7-a	100
J1-D	J2-I and J6-a	100
J1-G	J1-H, J3-E and P4-H	100
J1-A	J1-R and J6-F	100
J1-C	J3-I and J7-S	100
J1-E	J1-V and J3-O	100
J1-B	J3-K and J6-S	100
J1-d	J5-L	100
J1-b	J5-N	100
J1-M	J6-R	100
J1-Z	J2-H and J2-B	100
J2-D	J7-T, J8-E, J3-H, J8-G	100
J2-g	J6-T, J8-X, J3-L, J8-B	100
J2-A	J6-f, P2-C	100
J2-c	J7-f, P2-W	100
J1-J	None	100
J1-K	None	100
J1-F	J6-Y and J9-C	100
J1-S	J3-P and J7-G	100
J1-Y	J8-P	100
J2-V	J6-h	100
J2-S	None	100
J2-T	None	100
J2-j	None	100
J2-k	None	100

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J2-J	J5-F and J6-G	100
J2-O	J6-X	100
J2-M	J6-c	100
J2-C	J7-D	100
J2-R	J4-D, J5-D and J6-d	100
J2-N	J6-b	100
J2-h	J9-V	100
J2-P	J9-u	100
J2-d	J5-H and J7-E	100
J2-e	J4-B, J5-K and J7-d	100
J2-W	J7-l, J7-m and J9-G	100
J2-F	J6-l, J6-m and J9-L	100
J2-X	J7-j, J7-k and J9-l	100
J2-E	J6-j, J6-k and J9-D	100
J2-Z	J7-P, J7-N and J9-N	100
J2-L	J6-P, J6-N and J9-c	100
J2-Y	J9-K	100
J2-K	J9-B	100
J3-G	J8-A	100
J3-A	J8-F	100
J5-T	None	100
J5-P	None	100
J5-R	J6-D	100
P1-C	J5-C, P1-M, P1-V and P1-W	100
P2-D	J5-M, P2-L, P2-N and P2-P	100
P3-L	J6-U	100
J9-b	J8-U and J6-Z	100
P3-E	J6-e	100
J9-d	J6-H	100
J9-f	J6-W	100
P1-E	J6-B	100
P2-T	J8-K and J6-L	100
P2-S	J8-L and J6-M	100
P1-J	J6-C	100
P1-H	J6-g	100
P1-S	J6-A	100
P2-R	J8-H and J6-J	100
P2-K	J8-J and J6-K	100
P4-L	J7-e	100
J7-H	J9-A	100
J7-F	J9-F	100
P1-R	J7-A	100
J7-Z	J8-D and J9-k	100
P2-E	J8-Z and J7-J	100
P1-A	J7-g	100
J7-K	J8-a and P2-J	100
J7-L	J8-S and P2-H	100
J7-M	J8-T and P2-A	100
J7-c	J9-E	100
J7-W	J8-b and J9-j	100

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J7-B	P1-K	100
J7-C	P1-T	100
J7-U	P4-E	100
J7-Y	J8-R and P1-B	100
J9-M	None	100
J9-e	None	100
J9-H	P4-J	100
P1-X	P2-X	100
P2-B	P3-J	100
P3-X	P4-K	100

Table VII

Capacitor Check	Apply 150 ± 6 Volts rms at 60 Hz	All Other Pins to The Other Side of ac Source Except
C101	J7-g	J6-g, J6-A, J6-B, J7-c, J7-A, J7-B and J6-C
C201	J8-T and J7-M	J8-S, J7-L, J8-L, J6-M, J8-Z, J7-J, J8-K, J6-L, J8-H, J6-J, J8-J, J6-K, J8-a and J7-k.

4.4.3 Environmental testing (see 3.7).

4.4.3.1 Shock (spares only). The wiring harness relay assembly mounted in a cartridge assembly shall be connected to a shock producing device such as is specified in MIL-S-4456. Either the 150 pound test machine, as described therein, or a similar machine of capacity 150-400 pounds with calibrated tables shall be used. With the 150 pound machine, weights shall be added to the shock machine elevator such that the total weight on the elevator is 150 ± 1 pound. With the 150-400 pound machine, the calibrated tables shall be used in conjunction with the combined weight of the harness, relay assembly and mounting fixture to determine the machine setting to provide the proper impact. The wiring harness, relay assembly shall be shocked three times in each direction of the three principal mutually perpendicular axes. The shocks produced shall be impact shocks of 50 ± 1 g's with each shock impact having a time duration of 10 ± 1.5 milliseconds. The use of manufacturer's curves to determine amplitude and duration value listed above is acceptable provided the curves are in accordance with requirements of MIL-S-4456. After the shock tests, the wiring harness, relay assembly shall be tested as specified in 4.4.1 and 4.4.2.

4.4.3.2 Vibration (spares only). The wiring harness, relay assembly in a cartridge shall be mounted to a suitable vibration producing apparatus. The wiring harness, relay assembly shall be vibrated as specified in Table I in each of the three mutually perpendicular axes. During this test

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all the relay armatures in the cartridge assembly shall be unenergized except for K102 and K202 which shall be tested in both the energized (see Table VI) and unenergized conditions, and the contacts of all relays shall be monitored for contact chatter. The maximum allowable duration of contact chatter shall be 10.0 microseconds. Upon completion of the vibration test, the wiring harness, relay assembly shall be tested as specified in 4.4.1 and 4.4.2.

4.4.3.3 Altitude. The wiring harness, relay assembly shall be placed in a test chamber with the electrical circuits made externally available through appropriate electrical cabling. The internal temperature of the test chamber shall then be raised to +165 + 2°F. and held for a minimum period of 2 hours. Upon completion of the "soak" period and while continuing to hold the chamber temperature, the internal absolute pressure of the test chamber shall be reduced to 5.54 + 0.5 inches of mercury, (corresponding to an altitude of 40,000 feet). At the specified condition of the chamber pressure and temperature, the assembly shall comply with the requirements of 4.4.2. After meeting the requirements specified, the chamber temperature shall be uniformly reduced in temperature within a period of 2 hours to -65 + 2°F. The chamber absolute pressure of 5.54 inches of mercury shall not be hold as the chamber temperature is reduced. The -65 + 2°F. chamber temperature shall be held for a minimum period of 2 hours. The chamber absolute pressure shall then be reduced to 5.54 + 0.5 inches of mercury. After the specified absolute pressure has been obtained, the assembly shall comply with the requirement of 4.4.2. The chamber absolute pressure shall then be increased to ambient atmosphere and the chamber temperature increased uniformly within a period of 1 hour to between +68°F. and +95°F. Upon completion of the test, the assembly shall be tested as specified in 4.4.1 and 4.4.2.

5. PREPARATION FOR DELIVERY

5.1 Preservation and packaging. Preservation and packaging shall be level A or C as specified (see 6.1).

5.1.1 Level A. The harness shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.1.2 Level C. The harness shall be preserved and packaged in accordance with instructions contained in MIL-STD-1169.

5.2 Packing.

5.2.1 Levels A and C. The harness shall be packed in accordance with the applicable instructions contained in MIL-STD-1169.

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5.3 Marking. Marking shall be in accordance with MIL-STD-1169.

5.4 Data cards. Data cards shall be prepared in accordance with MIL-STD-1167.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

6.1.1 Procurement requirements.

a. Title, number, and date of this specification.

b. Selection of applicable level of packaging and packing required (see 5.1 and 5.2).

6.1.2 Contract data requirements (see 4.3.4.1). Data specified in 5.4 and 4.3.4.1 will be delivered as identified on a numbered DD Form 1664 when required on a DD Form 1423 incorporated in the contract.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

6.3 Acceptance inspection equipment (see 4.3.4.1). The contractor shall obtain approval from the Commander, Picatinny Arsenal, Dover, New Jersey 07801, ATTN: SARPA-QA-T.

6.4 Supersession data. This specification supersedes the requirements of Purchase Description PA-PD-1749 (Rev. 3) dated 8 May 1964.

Custodian:
Army - PA

Preparing activity:
Army - PA

Project No. 1135-A192

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

OMB Approval
No. 22-R255

INSTRUCTIONS: The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/ vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?
   A. GIVE PARAGRAPH NUMBER AND WORDING.

   B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

   ☐ YES ☐ NO *(If "Yes", in what way?)*

4. REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426 REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

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